"This Wasn't Made for Me": Recentering User Experience and Emotional Impact in the Evaluation of ASR Bias

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Studies on bias in Automatic Speech Recognition (ASR) tend to focus on reporting error rates for speakers of underrepresented dialects, yet less research examines the human side of system bias: how do system failures shape users' lived experiences, how do users feel about and react to them, and what emotional toll do these repeated failures exact? We conducted user experience studies across four U.S. locations (Atlanta, Gulf Coast, Miami Beach, and Tucson) representing distinct English dialect communities. Our findings reveal that most participants report technologies fail to consider their cultural backgrounds and require constant adjustment to achieve basic functionality. Despite these experiences, participants maintain high expectations for ASR performance and express strong willingness to contribute to model improvement. Qualitative analysis of open-ended narratives exposes the deeper costs of these failures. Participants report frustration, annoyance, and feelings of inadequacy, yet the emotional impact extends beyond momentary reactions. Participants recognize that systems were not designed for them, yet often internalize failures as personal inadequacy despite this critical awareness. They perform extensive invisible labor, including code-switching, hyper-articulation, and emotional management, to make failing systems functional. Meanwhile, their linguistic and cultural knowledge remains unrecognized by technologies that encode particular varieties as standard while rendering others marginal. These findings demonstrate that algorithmic fairness assessments based on accuracy metrics alone miss critical dimensions of harm: the emotional labor of managing repeated technological rejection, the cognitive burden of constant self-monitoring, and the psychological toll of feeling inadequate in one's native language variety. Understanding ASR bias requires attending not just to error rates but centering evaluation on both the experiences and emotional impacts borne by affected users.

CCS Concepts: • **Computing methodologies** → **Speech recognition**; • **Human-centered computing** → **Empirical studies in accessibility**; *User studies*; • **Social and professional topics** → *Race and ethnicity*.

Additional Key Words and Phrases: Automatic Speech Recognition, ASR, fairness, bias, user experience, emotional impacts, voice technology, African American English

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1 Introduction

Automatic Speech Recognition (ASR) systems have become deeply embedded in everyday technologies, from smartphones and smart speakers to customer service systems and accessibility tools. These systems promise hands-free convenience and natural interaction, yet mounting evidence demonstrates substantial performance disparities across

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dialects and accents. For speakers of non-mainstream language varieties, technical failures occur systematically and frequently, carrying consequences that extend far beyond mere inconvenience.

Prior research has documented these performance gaps extensively. Commercial ASR systems exhibit higher word error rates for certain groups of English speakers such as African American English speakers, Spanish-influenced English speakers, and Indigenous language speakers compared to speakers of dominant English varieties [15, 19, 28]. These disparities stem primarily from training data imbalances and linguistic features characteristic of non-dominant varieties that systems fail to accommodate [7, 22, 27]. These studies have established that bias exists and quantified its magnitude through error metrics, phonetic analyses, and controlled evaluations. These performance disparities reflect broader patterns of algorithmic discrimination documented across AI systems [1, 23].

Yet while the documentation of ASR bias continues to accumulate, significantly less attention has been paid to the human side of system bias. How do system failures shape users' lived experiences? How do users feel about and react to repeated misrecognition? What emotional toll do these failures exact over time? How do experiences vary across different dialect communities facing similar technical failures? And critically, how do these experiences shape users' relationships with voice technologies and their willingness to contribute to improving biased systems?

This study addresses these questions through a mixed-methods investigation of user experiences across four U.S. dialect communities: African American speakers in Atlanta, Georgia; Gulf Coast Creole and French speakers in Mississippi and Louisiana; Hispanic and Latine Caribbean speakers in Miami Beach, Florida, and Native American speakers in Tucson, Arizona. Through quantitative analysis of structured questionnaires and qualitative analysis of open-ended narrative responses, we move beyond performance metrics to examine the lived experiences and emotional impacts of algorithmic bias: the frustrations users articulate, the accommodations they make, their understandings of why systems fail them, and the complex decisions they face about whether to persist with, abandon, or contribute towards improving technologies that exclude them.

Our findings reveal that participants maintain high expectations for ASR performance and express overwhelming willingness to contribute to model improvement, yet most report that technologies fail to consider their cultural backgrounds and require constant adjustment to achieve basic functionality. Participants report frustration, annoyance, and feelings of inadequacy, yet the emotional impact extends beyond these momentary reactions. Qualitative analysis exposes deeper psychological costs: participants recognize that systems were not designed for them ("this wasn't made for me"), yet internalize failures as personal inadequacy despite this critical awareness. They perform extensive invisible labor, including code-switching, hyper-articulation, and emotional management, to make failing systems functional. Their linguistic and cultural knowledge remains unrecognized by technologies that encode particular varieties as standard while rendering others marginal. This tension between understanding exclusion and continuing to accommodate failing systems illuminates the complex situation of users navigating technologies that have become increasingly unavoidable yet remain systematically unreliable for their speech.

This research contributes to fairness and accountability scholarship by recentering user experience and emotional impact in evaluations of algorithmic bias. We demonstrate that fairness assessments based on accuracy metrics alone miss critical dimensions of harm: the emotional labor of managing repeated technological rejection, the cognitive burden of constant self-monitoring, and the psychological toll of feeling inadequate in one's native language. These invisible costs are central to experiences of algorithmic exclusion yet remain uncaptured in technical evaluations. We document how users interpret and respond to systematic technological failure, revealing sophisticated critical awareness alongside internalized self-blame, and exposing forms of harm that performance metrics fail to capture. Understanding ASR bias requires attending not just to error rates but to how bias is experienced and felt by affected users.

2 Related Work

Our work sits at the intersection of ASR bias analysis, user experience research on algorithmic harm, sociolinguistics, and critical frameworks for understanding linguistic justice in AI systems. We review three strands of research central to our research questions: (1) documented performance disparities and their causes, (2) user experiences of algorithmic bias and technological invisibility, and (3) participatory responses to biased systems and design justice.

2.1 ASR Performance Disparities Across Dialects

Studies have consistently documented performance disparities in commercial ASR systems across racial, ethnic, and dialectal lines. Koenecke et al. [15] found that major commercial systems from Amazon, Apple, Google, IBM, and Microsoft exhibited higher word error rates for African American speakers compared to white speakers, with disparities particularly pronounced for speakers of African American English (AAE). These disparities stem primarily from training data imbalances, with ASR models predominantly trained on Standard American English from speakers who are disproportionately white [6].

Recent work has expanded evaluation to multiple dialect varieties. Wassink et al. [28] analyzed ASR errors across four ethnicity-related sociolects from the Pacific Northwest, demonstrating differential error rates: Caucasian American English exhibited lowest error rates, followed by African American English, Yakama English, and ChicanX English. Harris et al. [12] evaluated ASR performance across four U.S. dialects (Standard American English, AAE, ChicanX English, and Spanglish), finding intersectional gender-dialect biases where systems performed worst for female speakers of minority dialects. Martin and Tang [19] showed how ASR systems fail to recognize the habitual "be" construction characteristic of AAE, a morphosyntactic feature with systematic grammatical function that systems misinterpret or ignore. Choe et al. [4] examined how speakers' native language phonologies predict specific categories of ASR errors across World Englishes. Approaches have also focused on different ways to develop systematic methods for measuring and analyzing ASR bias. For example, Feng et al. [8] analyzed specific phonemes to identify sources of bias, while Liu et al. [18] developed fairness measurement frameworks using demographically diverse datasets. However, critical analysis of this research literature itself reveals persistent conceptual limitations, including misconceptions about accent as something only certain speakers possess [25].

This body of work establishes that performance disparities are substantial, systematic, and attributable to specific phonological or morphosyntactic characterizing non-dominant varieties. However, these studies focus primarily on *documenting* failure rates rather than examining *user experiences* of failure or how communities respond when faced with persistent technological exclusion—the gaps our work addresses.

2.2 User Experiences of Algorithmic Bias

Emerging research examines how users experience algorithmic bias emotionally and psychologically. Mengesha et al. [20] investigated African American users' experiences with ASR errors through interviews and surveys, finding that participants perceived errors as culturally insensitive and felt the technology was not designed for them. Their qualitative work documented frustration, resignation, and perceptions of technological exclusion, providing crucial evidence that ASR bias creates psychological impacts beyond mere inconvenience. However, questions remain about how these feelings of exclusion interact with users' ongoing decisions to use, accommodate, or abandon failing technologies. Recent work has extended this analysis to AI voice services more broadly. Michel et al. [21] examined accent bias and

digital exclusion in synthetic AI voice services, finding parallel patterns of users feeling unrepresented by technologies that fail to accommodate their speech patterns.

Research on intersectional invisibility in organizational contexts provides a theoretical framework for understanding these experiences. Bhattacharyya and Berdahl [2] documented how women of color in workplaces experience systematic oversight and misrecognition at the intersection of multiple marginalized identities, identifying four forms of invisibility: erasure (being overlooked), homogenization (being confused with others), whitening (having cultural identity ignored), and exoticization (being treated as exotic other). Critically, these invisibility experiences create distinct affective reactions: ambiguous events (erasure, whitening) elicit shame and self-blame, while less ambiguous events (homogenization, exoticization) elicit anger directed at perpetrators. This dual pathway suggests that even when structural causes are visible, internalized responses may persist. This work builds on broader scholarship documenting how marginalized groups experience systematic misrecognition in institutional contexts, with implications for how users interpret technological failures [26].

We theorize that similar dynamics operate in human-technology interaction. When ASR systems consistently fail to recognize certain speakers, these failures may create *technological invisibility*—experiences of being unseen, misrecognized, or overlooked by systems. An open question is whether users maintain critical awareness that systems were not designed for them while simultaneously internalizing failure as personal inadequacy, and how these contradictory interpretations coexist in everyday technological practice.

2.3 Technological Labor and Participatory Tensions

When speakers of non-mainstream varieties must accommodate dominant norms to participate in technological systems, this accommodation constitutes a form of invisible labor: cognitive effort to modify natural speech, emotional management when systems fail, and temporal costs of repetition and troubleshooting [20]. This labor remains invisible to developers and privileged-dialect users for whom ASR "just works," yet represents a significant and inequitable burden for those whose speech falls outside training distributions. Pospisil et al. [24] find that about 76 percent of those who find the labor and concerns to be excessive will never return to the use of voice assistants. Our study operationalizes this concept through systematic documentation of coping strategies, examining what accommodations users report making and at what frequency across different dialect communities. The accommodation burden placed on speakers of non-dominant varieties reflects broader patterns of linguistic discrimination, wherein certain ways of speaking are systematically devalued while others are treated as neutral or standard [17].

Design justice frameworks argue that equitable technologies require centering marginalized communities throughout design processes [5]. Costanza-Chock articulates principles emphasizing that those most affected by design decisions should lead design processes, that communities hold expertise about their own experiences, and that design should dismantle rather than reproduce structural inequalities. Applied to ASR, these principles suggest that developing dialect-inclusive systems requires sustained partnership with speakers of underrepresented varieties, not merely collecting their voice data.

Yet tensions emerge when communities experiencing bias are asked to contribute to fixing biased systems. Harrington et al. [11] examine community-based collaborative design with Black and Brown communities, identifying tensions between researcher and community goals and emphasizing the importance of compensation, shared decision-making authority, and recognition of community expertise. When voice sample collection occurs without these protections, questions arise about equitable participation in AI development. Recent work has begun addressing these tensions in ASR contexts specifically. Hutchinson et al. [13] document participatory approaches to developing speech technologies

for Australian Aboriginal English, highlighting both opportunities and risks when developing ASR for marginalized language communities. A critical question emerges: if users recognize that ASR was not designed for them and experience repeated failures, what motivates continued willingness to participate in system improvement? This question sits at the intersection of critical awareness and ongoing technological engagement. Our study examines whether communities experiencing significant harm nevertheless remain willing to contribute to model improvement, and how this willingness relates to their experiences of exclusion and their understanding of why systems fail them.

2.4 Positioning This Work

This work reframes the study of ASR bias by shifting attention from system-centric performance metrics to the social and experiential dimensions of human–technology interaction. Rather than asking only how recognition accuracy varies across dialects, we ask how persistent misrecognition is understood, navigated, and emotionally processed by users whose speech falls outside dominant training distributions. By situating ASR failures within users' everyday practices, expectations, and interpretive frameworks, this study treats bias not solely as a technical property of models, but as a lived condition shaped by design choices, institutional norms, and unequal distributions of accommodation labor. In doing so, we bridge technical research on ASR disparities with human-centered and justice-oriented approaches to algorithmic accountability, emphasizing the importance of user experience as a central site for evaluating fairness in voice technologies.

3 Methodology

3.1 Research Design

Our study investigates user experiences with English Automatic Speech Recognition (ASR) technologies across four geographically and linguistically diverse communities in the United States. Our methods combine structured questionnaires with open-ended responses to capture both quantitative patterns and qualitative experiences of ASR system failures and their psycho-emotional impacts.

Data collection occurred between July, 2023 and August, 2025 for four sites: Atlanta, Georgia; the Gulf Coast region (Mississippi and Louisiana); Miami Beach, Florida; and Tucson, Arizona. These communities represent systematically underrepresented varieties with well-documented linguistic structures distinct from Standard American English [3, 9, 10, 14, 16].

3.2 Participants

We recruited a total of 494 participants across the four sites; of these, 215 of them, which forms the basis of this study, met our inclusion criteria (target linguistic and demographic background) and provided near-complete survey responses. All of the participants self-identified as speakers of underrepresented dialects in the United States and reported regular use of voice-enabled technologies. Recruitment strategies varied by site to reach distinct dialect communities: Atlanta participants (n=86) were recruited at local community centers and a conference of black engineers; Gulf Coast participants (n=61) were recruited from the community by word-of-mouth; Miami Beach participants (n=43) were recruited from the student and staff body at Florida International University; and Tucson participants (n=25) were recruited through the Pascua Yaqui Cultural Center and the intertribal Native American Student Affairs center at the University of Arizona.

Table 1 presents an overview of the demographic composition of the participants. The Atlanta sample (n=86) consisted primarily of Black/African American participants from the Southeast. The Gulf Coast sample (n=61) included Black/African American participants and French Creole speakers from Mississippi and Louisiana. The Miami Beach sample (n=43) included Caribbean Hispanic/Latine participants, Black Caribbean, and Black/African American participants from Florida. The Tucson sample (n=25) consisted primarily of Native American participants from Arizona, including speakers of Yaqui, Navajo, Tohono O'odham, and other Indigenous backgrounds.

Table 1. Participant Demographics by Research Site

Characteristic	**Atlanta** (n=86)	**Gulf Coast** (n=61)	**Miami Beach** (n=43)	**Tucson** (n=25)
Gender Identity				
Women	48 (56%)	20 (33%)	31 (72%)	19 (76%)
Men	37 (43%)	38 (62%)	11 (26%)	6 (24%)
Non-binary	1 (1%)	1 (2%)	0 (0%)	0 (0%)
Trans woman	0 (0%)	1 (2%)	0 (0%)	0 (0%)
Prefer not to state	0 (0%)	1 (2%)	1 (2%)	0 (0%)
Age (years)				
Range	19–60	18–68	18–63	18–54
Mean (SD)	29.9 (8.5)	33.6 (12.1)	23.4 (11.2)	25.8 (10.2)

3.3 Data Collection

3.3.1 Questionnaire Design. We developed a questionnaire investigating participants' experiences with voice-enabled technologies, their perceptions of ASR system performance, and the psycho-emotional impacts of recognition failures. The questionnaire included both structured multiple-choice items and open-ended narrative prompts.

Key questionnaire topics included:

- Demographic information (age, location, ethnicity, gender identity)
- Self-description of speech characteristics (Q1) and speech challenges (Q4)
- Types of voice-enabled technologies used (Q5–Q7) and purposes of use (Q6a, Q8a)
- Frequency of use for different purposes (Q12a–f: 5-point Likert scale)
- Discontinued use and reasons (Q10: open-ended)
- Memorable experiences with ASR technologies (Q11: open-ended)
- Specific instances of dialect or accent-related recognition failures (Q13: open-ended)
- Types of challenges experienced (Q14: multiple-choice with options including cultural/ethnic/racial exclusion, contextual misunderstanding, physical effort requirements, self-consciousness)
- Coping strategies employed when systems fail (Q15a: multiple-choice with options including repetition, speech modification, enunciation, speaking louder/slower, imitation, manual task completion, asking others for help, technology abandonment)
- Emotional responses to system failures (Q16: open-ended)
- Expectations for ASR accuracy (Q17: 5-point scale from "Always [100%]" to "Never [0–20%]")
- Willingness to provide voice samples for system improvement (Q18a: Yes/Maybe/No)

- Willingness to pay for improved services (Q19: multiple payment tier options including compensation preference)

The full questionnaire can be found in the appendix. The Atlanta collection was the first to occur. Afterward, the interview and survey were modified slightly to increase the likelihood that question wording would elicit the information of interest. We will mention below where the Atlanta questionnaire differed slightly from the other locations' surveys.

3.3.2 Data Collection Procedures. Two procedures were used for the collection of data. First, a semi-structured interview was delivered in-person and audio-recorded. Following consenting procedures, respondents were invited to respond verbally to an interviewer who delivered all questions orally. Second, we offered the choice to all respondents to participate online via a self-paced survey created using the JotForm platform (jotform.com). The survey included radio-button, multiple-choice, short-form text entry and long-form text entry as well as audio widget response options, as appropriate to the question. All participants, regardless of collection method, responded to the same questionnaire, enabling cross-site comparison for both quantitative and qualitative analyses.

3.4 Analytical Approach

3.4.1 Quantitative Analysis. We report descriptive statistics (frequencies, percentages) for all structured questionnaire items to characterize response patterns within and across sites. For multiple-selection items (Q14: challenges experienced; Q15a: coping strategies employed), participants could select all applicable options, resulting in non-mutually exclusive categories. Percentages therefore reflect the proportion of participants who endorsed each option and may sum to greater than 100%. Comparisons across sites are reported descriptively given the non-probability sampling approach and the distinct demographic characteristics of each dialect community. Where notable differences emerged between sites, we interpret these in light of the specific linguistic and cultural contexts of each community.

3.4.2 Qualitative Analysis. We conducted qualitative analysis of open-ended responses to three key questions: memorable experiences with ASR technologies (Q11), specific instances of dialect-related recognition challenges (Q13), and emotional responses to system failures (Q16). Several major themes emerged from this analysis, representing distinct dimensions of how participants experience and respond to ASR bias. These themes capture both surface-level impacts (frustration with errors, need for repetition) and deeper psychological dimensions (internalized blame despite critical awareness of exclusion, feelings of linguistic and cultural invisibility).

Participant quotes are identified by site and unique participant ID (e.g., [Atlanta] CD24-055) to enable verification while protecting anonymity. When quotes contained implicit identifying information, minor details were altered to preserve confidentiality without changing the meaning substantively.

4 Results

We present findings below from our investigation of user experiences with ASR technologies. Since some participants could choose to skip some questions, response rates varied by question. Therefore, sample sizes differ slightly across questions and are reported accordingly in the respective tables.

4.1 Quantitative Findings

4.1.1 Types of Challenges Experienced. Participants identified multiple types of challenges when using ASR technologies (Table 2). Cultural and ethnic exclusion was reported by a majority of participants overall (53.5%), with notably higher

rates among Tucson participants (72.0%) and Miami Beach participants (62.8%) compared to Gulf Coast (52.5%) and Atlanta (44.2%) participants.

Contextual misunderstanding—voice assistants taking speech out of context or performing unintended actions—was reported by 51.2% of participants overall, with 76.0% of Tucson participants and 69.8% of Miami Beach participants reporting the issue, compared to 47.5% of Gulf Coast and 37.2% of Atlanta participants.

Physical effort requirements showed a similar pattern, with Tucson (60.0%) and Miami Beach (39.5%) participants reporting higher rates than Gulf Coast (37.7%) and Atlanta (27.9%) participants. Self-consciousness about speech when interacting with voice technologies was reported by 44.0% of Tucson participants, compared to 36.1% of Gulf Coast, 25.6% of Miami Beach, and 17.4% of Atlanta participants.

Table 2. Types of Challenges Experienced by Site (Q14; Multiple Selections Allowed)

	Atlanta (n=86)		Gulf Coast (n=61)		Miami Beach (n=43)		Tucson (n=25)		Total (N=215)	
Challenge Type	n	%	n	%	n	%	n	%	n	%
Cultural/ethnic/racial exclusion	38	44.2	32	52.5	27	62.8	18	72.0	115	53.5
Contextual misunderstanding	32	37.2	29	47.5	30	69.8	19	76.0	110	51.2
Physical effort required	24	27.9	23	37.7	17	39.5	15	60.0	79	36.7
Self-conscious about speech	15	17.4	22	36.1	11	25.6	11	44.0	59	27.4

4.1.2 Coping Strategies. When ASR systems failed to recognize their speech, participants reported employing a variety of coping strategies (Table 3). Repetition was the most common strategy overall (82.8%), with consistently high rates across all sites ranging from 75.4% in the Gulf Coast to 87.2% in Atlanta.

Speech modification was a prevalent coping mechanism. In Atlanta, where participants were offered a more open-ended response option, 58.1% reported changing the way they speak or code-switching. At other sites, where more granular options were available, participants reported slowing down speech (Gulf Coast 68.9%, Miami Beach 72.1%, Tucson 76.0%), speaking louder (Gulf Coast 59.0%, Miami Beach 83.7%, Tucson 88.0%), enunciating more (Gulf Coast 47.5%, Miami Beach 83.7%, Tucson 84.0%), and changing words or phrases (Gulf Coast 54.1%, Miami Beach 60.5%, Tucson 72.0%).

Technology abandonment strategies were also prevalent. Complete discontinuation of ASR use was reported by 49.8% of participants overall, with Tucson showing the highest rate (80.0%), followed by Miami Beach (55.8%), Atlanta (47.7%), and Gulf Coast (36.1%). Manual task completion was common in the Gulf Coast (59.0%), Miami Beach (76.7%), and Tucson (80.0%). Social workarounds included imitating the voice assistant's speech patterns (26.5% overall) and asking another person to operate the device (21.9% overall).

4.1.3 Expectations for ASR Performance. Most participants expected ASR systems to understand their speech correctly at least 75% of the time (Table 4). Across all sites, 53.7% of participants expected ASR to work "often" (75% of the time), and 27.6% expected it to work "always" (100% of the time). Only 15.0% expected ASR to work only "sometimes" (50%), and 3.7% expected it to work "rarely" (25%) or never. These high expectations—with over 80% anticipating at least 75% accuracy—contrast sharply with the challenges and coping strategies reported above, suggesting a substantial gap between user expectations and actual ASR performance for these dialect communities.

Table 3. Coping Strategies Employed by Site (Q15a; Multiple Selections Allowed)

	Atlanta (n=86)		Gulf Coast (n=61)		Miami Beach (n=43)		Tucson (n=25)		Total (N=215)	
Strategy	n	%	n	%	n	%	n	%	n	%
Repeat utterance	75	87.2	46	75.4	36	83.7	21	84.0	178	82.8
Code-switch/modify speech[a]	50	58.1	–	–	–	–	–	–	–	–
Change words/phrases[b]	–	–	33	54.1	26	60.5	18	72.0	77	59.7
Slow down speech[b]	–	–	42	68.9	31	72.1	19	76.0	92	71.3
Speak louder[b]	–	–	36	59.0	36	83.7	22	88.0	94	72.9
Enunciate more[b]	–	–	29	47.5	36	83.7	21	84.0	86	66.7
Stop using technology	41	47.7	22	36.1	24	55.8	20	80.0	107	49.8
Complete task manually[b]	–	–	36	59.0	33	76.7	20	80.0	89	69.0
Imitate voice assistant	14	16.3	16	26.2	20	46.5	7	28.0	57	26.5
Ask another person	9	10.5	13	21.3	16	37.2	9	36.0	47	21.9

[a]Atlanta used an earlier survey version with this broader category instead of granular speech modification options.
[b]Option not available for Atlanta; percentage calculated from Gulf Coast, Miami Beach, and Tucson only (n=129).

Table 4. Expectations for ASR Accuracy by Site (Q17)

	Atlanta (n=85)		Gulf Coast (n=61)		Miami Beach (n=43)		Tucson (n=25)		Total (N=214)	
Expected Accuracy	n	%	n	%	n	%	n	%	n	%
Always (100%)	28	32.9	15	24.6	9	20.9	7	28.0	59	27.6
Often (75%)	43	50.6	32	52.5	28	65.1	12	48.0	115	53.7
Sometimes (50%)	12	14.1	11	18.0	4	9.3	5	20.0	32	15.0
Rarely (25%)	2	2.4	3	4.9	1	2.3	1	4.0	7	3.3
Never (0–20%)	0	0.0	0	0.0	1	2.3	0	0.0	1	0.5

4.1.4 Willingness to Contribute Voice Samples. Despite experiencing significant challenges with ASR systems, a substantial majority of participants (75.5%) expressed willingness to contribute voice samples to help improve recognition accuracy for diverse dialects (Table 5). An additional 17.9% indicated they might contribute, while only 6.6% declined. Willingness was highest in Miami Beach (88.1%) and Gulf Coast (85.2%), followed by Tucson (76.0%) and Atlanta (61.9%).

Table 5. Willingness to Provide Voice Samples by Site (Q18a)

	Atlanta (n=84)		Gulf Coast (n=61)		Miami Beach (n=42)		Tucson (n=25)		Total (N=212)	
Response	n	%	n	%	n	%	n	%	n	%
Yes	52	61.9	52	85.2	37	88.1	19	76.0	160	75.5
Maybe	26	31.0	6	9.8	2	4.8	4	16.0	38	17.9
No	6	7.1	3	4.9	3	7.1	2	8.0	14	6.6

4.1.5 Willingness to Pay for Improved Services. A majority of participants (63.3%) expressed willingness to pay for improved ASR services that better recognize diverse dialects (Table 6). Gulf Coast participants showed the highest

willingness (81.5%), followed by Atlanta (58.8%), Tucson (56.0%), and Miami Beach (53.5%). Among those willing to pay, preferences also varied by site.

Table 6. Willingness to Pay for Improved ASR Services by Site (Q19)

	Atlanta (n=85)		Gulf Coast (n=54)		Miami Beach (n=43)		Tucson (n=25)		Total (N=207)	
Response	n	%	n	%	n	%	n	%	n	%
Prefer 1-time payment	26	30.6	10	18.5	13	30.2	5	20.0	54	26.1
< $10/month	15	17.6	6	11.1	5	11.6	6	24.0	32	15.5
$10–20/month	4	4.7	13	24.1	4	9.3	2	8.0	23	11.1
$20–30/month	3	3.5	12	22.2	1	2.3	1	4.0	17	8.2
> $30/month	2	2.4	3	5.6	0	0.0	0	0.0	5	2.4
Not willing to pay	35	41.2	10	18.5	20	46.5	11	44.0	76	36.7

4.2 Qualitative Findings: Lived Experiences Across Four Communities

Complementing the quantitative results, we present analysis of open-ended narratives which revealed recurring patterns in how users interpret, respond to, and make sense of ASR failures. Rather than simply expressing frustration, participants' accounts exposed the sophisticated strategies they employ to navigate exclusionary systems, the emotional labor required to maintain relationships with failing technologies, and the critical awareness they hold about why these systems fail them.

4.2.1 Recognition of Exclusionary Design. Across all sites, participants articulated explicit awareness that voice technologies were not designed with their communities in mind. This recognition manifested not as speculation but as lived certainty derived from repeated failures. A Navajo speaker from Tucson captured this sentiment directly: "Because Navajo is my first language ... I get this automatic feeling of inferiority if that makes sense. Because ... in my mind I'm thinking this is not something that was made for me automatically. And I'm stepping into a world of what other people are doing and not me because I'm native." (Tucson CD24-606). This participant's elaboration reveals how exclusion becomes internalized through accumulated interactions with systems that consistently fail to recognize one's voice.

A Gulf Coast participant made the systemic nature of this exclusion explicit: "It's gonna be for everybody and that it needs to be based on everybody's speech and everybody's culture. But I don't take stuff that deep. You know I just look at stuff different, like I know that places weren't created for everyone. Yeah most things aren't inclusive" (Gulf Coast CD24-546). This statement reveals a paradox: the participant demonstrates sophisticated critical analysis—recognizing that technologies marketed as universal exclude their community—while simultaneously describing this exclusion as unremarkable, something not worth dwelling on because it reflects a broader pattern of non-inclusive design.

An Atlanta participant made the racial dimensions of this exclusion explicit: "It is very frustrating because for technologies designed to make life better don't work for bi-lingual people of color. Whereas other non-[people of color] around me have no trouble with the technology" (Atlanta CD24-188). This comparative observation highlights how exclusion becomes visible through contrast with others' seamless experiences.

These statements demonstrate that users do not simply experience ASR failures as random technical glitches. Rather, they understand these failures as evidence of systematic exclusion from the design process, reflecting broader patterns of whose voices, languages, and ways of speaking are centered in technological development.

4.2.2 Internalized Blame and Emotional Burden. Despite recognizing systemic exclusion, many participants described internalizing ASR failures as personal inadequacy. This pattern was particularly striking given participants' simultaneous awareness that the problem was not theirs to solve. A Tucson participant articulated this tension: "It's my problem that this thing's not working. Never. My thought is never like there's something wrong with this you know. Like this needs to be fixed or this is not perfect you know. That thought never entered enters my mind. I'm always like what am I doing wrong. How can I say it to make this thing work you know. And so I think that that's the part that gets most people. Yeah it's kind of like what did I do wrong when it's really not your fault" (Tucson CD24-606).

This participant's account reveals the psychological process of misplaced responsibility: despite knowing that the system is flawed, the immediate emotional response is self-blame. The phrase "what am I doing wrong when it's really not your fault" captures the gap between the critical understanding and the unavoidable emotional experience.

Another Tucson participant described feelings of isolation: "It kinda makes you frustrated actually. ... Like I said, most people don't have problem with it, you know. Like why am I the only one struggling with this" (Tucson CD24-611). This statement illustrates how ASR failures create the illusion of individual inadequacy in addition to collective exclusion, particularly when failures are not publicly visible or discussed.

The most succinct expression of this emotional burden came from an Atlanta participant: "I feel inadequate speaking my own first language" (Atlanta CD24-198). This statement captures a profound form of linguistic alienation—the internalized sense that one's native language variety, the language of intimate relationships and cultural identity, is somehow deficient because it fails to conform to the narrow linguistic models encoded in voice recognition systems.

4.2.3 Strategic Accommodation to System Failure. Participants across all sites described extensive behavioral modifications required to achieve even basic functionality with voice technologies. These adaptations ranged from phonetic adjustments to code-switching, representing invisible labor that users perform to accommodate systems that do not accommodate them.

A Gulf Coast participant described the mental and linguistic accommodation required: "I had to adjust my pronunciation, speaking more slowly and deliberately, almost like I was 'talking to a foreigner.' It worked, but it felt awkward, like I was compromising my natural way of speaking to accommodate the technology" (Gulf Coast CD24-425). The comparison to "talking to a foreigner" reverses the expected relationship—rather than the technology adapting to diverse users, users must treat the technology as a non-native speaker requiring careful and modified input.

Another Gulf Coast participant made the racialized nature of this accommodation explicit: "Yeah, I put on my white voice when I use it. Like I put on my voice when I talk to Siri, like when I want it to be immediate" (Gulf Coast CD24-546). This participant's casual reference to deploying a "white voice" reveals both the normalization of code-switching as necessary for technological interaction and the explicit awareness that the system recognizes certain racialized varieties of English while failing to recognize others.

An Atlanta participant expressed the desire for authenticity denied by these constant accommodations: "Every time I use my natural dialect I have problems. I have learned to adjust my dialect for voice recognition systems. I wish I could just speak normally" (Atlanta CD24-055). The phrase "speak normally" highlights how technological failures redefine speakers' natural, authentic speech as abnormal or problematic.

A Tucson participant described the cognitive burden of constant self-monitoring: "I find myself thinking a lot more about how I'm talking than I would in a daily day to day life" (Tucson CD24-629). This statement reveals how voice technologies introduce extra cognitive load, forcing users to monitor and adjust their speech in real-time rather than focusing on communicative goals.

These accounts demonstrate that what technical evaluations measure as “minor performance degradation” translates into substantial ongoing labor for users, who must constantly adjust their linguistic behavior to achieve functionality that other users access seamlessly.

4.2.4 Cultural and Linguistic Exclusion. Participants described specific ways their linguistic and cultural knowledge remains unrecognized by voice technologies, requiring either abandonment of these resources or creation of workarounds. A participant of Guyanese heritage in Miami Beach proposed a solution: "We need a Caribbean Alexa... Because you know in the Caribbean we have our own Creole. And so an Alexa Caribbean is going to know the Caribbean Creole because if you ask Alexa something that is Caribbean based she doesn’t have that culture nor ethnicity that she can relate to" (Miami Beach CD24-569). This participant’s proposal anthropomorphizes the problem effectively—the system lacks not just linguistic data but cultural knowledge and context necessary for genuine comprehension.

A Miami Beach participant described how accent becomes marked as non-normative: “When I call someone I have to use non-Miami accent, normal people talk” (Miami Beach CD24-600). The equation of “non-Miami accent” with “normal people talk” reveals how ASR systems encode particular varieties as standard while rendering others abnormal, creating a hierarchy of acceptable speech.

Another Gulf Coast participant identified the technical gap: “Siri doesn’t take into account different dialects (ex. AAVE, Ebonics, French accents, slang)” (Gulf Coast CD24-503), naming specific varieties and features excluded from the system’s model of language.

A Gulf Coast participant described the specific linguistic features that remain unrecognized: “Voice to text doesn’t recognize or know what to do with common slang terms like ’chile.’ I have had issues where voice to text doesn’t understand when I am trying to play a song that has a title that is slang. If I do not speak super proper Siri doesn’t understand me. Most of my names with black cultural roots aren’t understood when I ask Siri to call them so Siri will call the wrong person” (Gulf Coast CD24-515). This account catalogs the cumulative exclusions—slang terms, song titles reflecting cultural tastes, personal names—that together render a person’s linguistic and social world invisible to the system.

4.2.5 Critical Awareness of Systemic Failures. Many participants demonstrated sophisticated understanding of why systems fail them, locating problems in inadequate training data, narrow linguistic models, or design processes that exclude diverse communities. A Gulf Coast participant analyzed the technical limitation: “I will realize that it is not intelligent enough, and it needs to constantly improve and update the language habits and database” (Gulf Coast CD24-495). This participant identifies insufficient training data as the source of failures rather than accepting user blame.

A Gulf Coast participant articulated the disconnect between system capabilities and user needs: “I feel a sense of disconnection from the technology, as if it’s not designed to accommodate my way of speaking” (Gulf Coast CD24-427). Although the statement suggests hypothesis rather than certainty, the participant’s experience provides strong evidence for this conclusion.

Some participants contextualized ASR failures within broader technological exclusion. The Gulf Coast participant who noted “It’s gonna be for everybody but it’s not based on everybody” demonstrated understanding that claimed universality often masks particular, exclusionary design choices (Gulf Coast CD24-546).

This critical awareness did not necessarily translate into expectations that systems would not change. As one participant noted: “I know that it will get better” (Gulf Coast CD24-537), expressing faith in future improvement that allows acceptance of present failures. Another stated: “it’s easy to feel frustrated but sometimes I just have to tell myself

to, like you know, relax, you know, it's learning" (Miami Beach CD24-558), framing the persistent failures as temporary growing pains rather than unsolvable problems.

5 Discussion

5.1 The Expectations-Experience Gap

Over 80% of participants expected ASR systems to understand their speech at least 75% of the time, yet more than half reported that technologies fail to consider their cultural backgrounds. This gap between expectations and experience has significant implications. Marketing narratives tend to promise "natural" voice interaction, creating expectations that systems will accommodate linguistic diversity. When systems fail, users who have internalized these promises may attribute failures to their own speech rather than to system limitations. The participant who described always thinking "what am I doing wrong" despite knowing "it's really not your fault" (Tucson CD24-606) captures this dynamic precisely.

This expectations gap also explains the substantial rates of technology abandonment we observed. When systems consistently fail to meet reasonable expectations, users face a choice between continued accommodation labor and discontinuation. That 80.0% of Tucson participants and 55.8% of Miami Beach participants reported stopping use of ASR technologies suggests that for many users, the costs of accommodation eventually outweigh the benefits of use.

5.2 The Paradox of Critical Awareness and Internalized Blame

One of the most significant findings is the coexistence of sophisticated critical awareness with internalized self-blame. Participants articulated clear understanding that systems were "not made for" them, that training data excludes their communities, and that their failures reflect design choices rather than personal deficiencies. Yet this critical awareness did not prevent emotional responses of inadequacy, frustration directed inward, and self-monitoring of speech.

This paradox extends theoretical work on intersectional invisibility [2] into human-technology interaction. Bhattacharyya and Berdahl found that ambiguous invisibility experiences (erasure, whitening) elicit shame and self-blame even when structural causes are recognized. Our findings suggest similar dynamics operate with ASR: repeated system failures create cumulative experiences of technological invisibility that generate internalized responses despite intellectual understanding of systemic causes. The participant who described feeling "inadequate speaking my own first language" (Atlanta CD24-198) exemplifies how technological exclusion can create profound linguistic alienation.

5.3 Invisible Labor and Unequal Burden

The coping strategies participants reported—repetition (82.8%), slowing speech (71.3%), speaking louder (72.9%), enunciating more (66.7%)—represent substantial invisible labor. This labor is invisible in two senses: it goes unrecognized in technical evaluations that measure only accuracy, and it remains unseen by users for whom systems "just work."

The participant who described deploying a "white voice" when using Siri (Gulf Coast CD24-600) makes explicit what remains implicit in aggregate statistics: accommodation to ASR systems requires performing linguistic identities that align with training data while suppressing natural speech patterns. This linguistic accommodation represents a form of everyday discrimination, where access to technology requires abandoning authentic self-expression [17].

Notably, this burden falls unequally even among marginalized communities. Tucson participants (primarily Native American speakers) reported the highest rates of challenges across all categories and the highest rates of technology abandonment (80.0%), suggesting that speakers of Indigenous languages and varieties who rarely feature in accessibility and fairness scholarship face particularly severe exclusion from voice technologies.

5.4 Implications for Participatory System Development

The finding that 75.5% of participants expressed willingness to contribute voice samples despite experiencing significant harm raises important questions for participatory approaches to system development. This willingness might reflect genuine desire to improve systems for future users, pragmatic recognition that improvement requires community participation, or resignation to contributing labor to systems that have already extracted accommodation costs. Simultaneously, a majority of participants (63.3%) were willing to pay for improved services. These findings support calls for design justice frameworks [5] that center affected communities not merely as data sources but as partners with authority over development processes.

6 Conclusions

This study demonstrates that understanding ASR bias requires moving beyond accuracy metrics to examine lived experiences of technological exclusion. Our findings reveal that critical awareness of systemic bias coexists with feelings of invisibility and internalized blame: participants who clearly articulated that systems were not made for them nevertheless described feeling inadequate speaking their native languages. This paradox, knowing the system is at fault while still blaming oneself, suggests that intellectual understanding provides incomplete protection against the psychological toll of repeated technological rejection.

The accommodation strategies participants reported represent substantial invisible labor that current fairness metrics fail to capture. Code-switching, hyper-articulation, repetition, and constant self-monitoring constitute an unequal tax on users whose speech falls outside narrow training distributions. Access to voice technology is often contingent on suppressing authentic linguistic identity. This burden falls most heavily on already marginalized communities.

Our study contributes to growing recognition that technical fairness metrics alone cannot capture the full dimensions of algorithmic harm. Voice technologies create worlds that include some speakers while excluding others, and the invisible costs of exclusion extend far beyond what error rates can measure. Yet our findings also reveal grounds for optimism: despite experiencing significant harm, participants overwhelmingly expressed willingness to contribute to model improvement and pay for improved services. This finding suggests that at least some in affected communities remain invested in shaping voice technologies rather than abandoning them entirely. Realizing this potential requires moving beyond extraction-based approaches that treat marginalized speakers as data sources toward participatory frameworks that recognize community expertise, ensure equitable compensation, and grant meaningful authority over how systems are developed and deployed. The path toward inclusive voice technologies runs through the communities currently excluded from them.

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References

[1] Ruha Benjamin. 2019. *Race After Technology: Abolitionist Tools for the New Jim Code.* Wiley. Google-Books-ID: nPy9uwEACAAJ.

[2] Barnini Bhattacharyya and Jennifer L. Berdahl. 2023. Do you see me? An inductive examination of differences between women of color's experiences of and responses to invisibility at work. *Journal of Applied Psychology* 108, 7 (2023), 1073–1095. doi:10.1037/apl0001072 Place: US Publisher: American Psychological Association.

[3] Su Lin Blodgett, Lisa Green, and Brendan O'Connor. 2016. Demographic Dialectal Variation in Social Media: A Case Study of African-American English. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Jian Su, Kevin Duh, and Xavier Carreras (Eds.). Association for Computational Linguistics, Austin, Texas, 1119–1130. doi:10.18653/v1/D16-1120

[4] June Choe, Yiran Chen, May Pik Yu Chan, Aini Li, Xin Gao, and Nicole Holliday. 2022. Language-specific Effects on Automatic Speech Recognition Errors for World Englishes. In *Proceedings of the 29th International Conference on Computational Linguistics*. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, 7177–7186. https://aclanthology.org/2022.coling-1.628

[5] Sasha Costanza-Chock. 2020. *Design Justice: Community-Led Practices to Build the Worlds We Need.* The MIT Press. doi:10.7551/mitpress/12255.001.0001

[6] Rachel Dorn. 2019. Dialect-Specific Models for Automatic Speech Recognition of African American Vernacular English. In *Proceedings of the Student Research Workshop Associated with RANLP 2019*, Venelin Kovatchev, Irina Temnikova, Branislava Šandrih, and Ivelina Nikolova (Eds.). INCOMA Ltd., Varna, Bulgaria, 16–20. doi:10.26615/issn.2603-2821.2019_003

[7] Siyuan Feng, Bence Mark Halpern, Olya Kudina, and Odette Scharenborg. 2024. Towards inclusive automatic speech recognition. *Computer Speech & Language* 84 (March 2024), 101567. doi:10.1016/j.csl.2023.101567

[8] Siyuan Feng, Olya Kudina, Bence Mark Halpern, and Odette Scharenborg. 2021. Quantifying Bias in Automatic Speech Recognition. doi:10.48550/arXiv.2103.15122 arXiv:2103.15122 [cs, eess].

[9] Carmen Fought. 2003. *Chicano English in Context.* Palgrave Macmillan. Google-Books-ID: AzskWeEaWrEC.

[10] Lisa J. Green. 2002. *African American English: A Linguistic Introduction.* Cambridge University Press, Cambridge. doi:10.1017/CBO9780511800306

[11] Christina Harrington, Sheena Erete, and Anne Marie Piper. 2019. Deconstructing Community-Based Collaborative Design: Towards More Equitable Participatory Design Engagements. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW (Nov. 2019), 216:1–216:25. doi:10.1145/3359318

[12] Camille Harris, Chijioke Mgbahurike, Neha Kumar, and Diyi Yang. 2024. Modeling Gender and Dialect Bias in Automatic Speech Recognition. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 15166–15184. doi:10.18653/v1/2024.findings-emnlp.890

[13] Ben Hutchinson, Celeste Rodríguez Louro, Glenys Collard, and Ned Cooper. 2025. Designing Speech Technologies for Australian Aboriginal English: Opportunities, Risks and Participation. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*. Association for Computing Machinery, New York, NY, USA, 108–124. doi:10.1145/3715275.3732010

[14] Thomas Klingler. 2003. *If I Could Turn My Tongue Like That: The Creole Language of Pointe Coupee Parish, Louisiana.* LSU Press. Google-Books-ID: Q4B2kWdViU4C.

[15] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R. Rickford, Dan Jurafsky, and Sharad Goel. 2020. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences* 117, 14 (April 2020), 7684–7689. doi:10.1073/pnas.1915768117 Publisher: Proceedings of the National Academy of Sciences.

[16] William Leap. 1993. *American Indian English.* University of Utah Press. Google-Books-ID: pL55AAAAIAAJ.

[17] Rosina Lippi-Green. 2012. *English with an Accent: Language, Ideology and Discrimination in the United States* (2 ed.). Routledge, London. doi:10.4324/9780203348802

[18] Chunxi Liu, Michael Picheny, Leda Sarı, Pooja Chitkara, Alex Xiao, Xiaohui Zhang, Mark Chou, Andres Alvarado, Caner Hazirbas, and Yatharth Saraf. 2021. Towards Measuring Fairness in Speech Recognition: Casual Conversations Dataset Transcriptions. doi:10.48550/arXiv.2111.09983 arXiv:2111.09983 [cs, eess].

[19] Joshua L. Martin and Kevin Tang. 2020. Understanding Racial Disparities in Automatic Speech Recognition: The Case of Habitual "be". In *Interspeech 2020*. ISCA, 626–630. doi:10.21437/Interspeech.2020-2893

[20] Zion Mengesha, Courtney Heldreth, Michal Lahav, Juliana Sublewski, and Elyse Tuennerman. 2021. "I don't Think These Devices are Very Culturally Sensitive."—Impact of Automated Speech Recognition Errors on African Americans. *Frontiers in Artificial Intelligence* 4 (Nov. 2021). doi:10.3389/frai.2021.725911 Publisher: Frontiers.

[21] Shira Michel, Sufi Kaur, Sarah Elizabeth Gillespie, Jeffrey Gleason, Christo Wilson, and Avijit Ghosh. 2025. "It's not a representation of me": Examining Accent Bias and Digital Exclusion in Synthetic AI Voice Services. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*. ACM, Athens Greece, 228–245. doi:10.1145/3715275.3732018

[22] Mikel K. Ngueajio and Gloria Washington. 2022. Hey ASR System! Why Aren't You More Inclusive? Automatic Speech Recognition Systems' Bias and Proposed Bias Mitigation Techniques. A Literature Review. Vol. 13518. 421–440. doi:10.1007/978-3-031-21707-4_30 arXiv:2211.09511 [cs, eess].

[23] Safiya Umoja Noble. 2018. *Algorithms of Oppression: How Search Engines Reinforce Racism.* NYU Press.

[24] Bettina Pospisil, Thilo Sauter, Albert Treytl, Edith Huber, and Walter Seböck. 2022. "Totally Unnecessary" or "Simply Convenient" – About Users and Non-Users of Voice Assistants. In *2022 15th International Conference on Human System Interaction (HSI)*. 1–7. doi:10.1109/HSI55341.2022.9869441 ISSN: 2158-2254.

[25] Kerri Prinos, Neal Patwari, and Cathleen A. Power. 2024. Speaking of accent: A content analysis of accent misconceptions in ASR research. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*. ACM, Rio de Janeiro Brazil, 1245–1254. doi:10.1145/3630106.3658969

[26] John R. Rickford and Sharese King. 2016. Language and linguistics on trial: Hearing Rachel Jeantel (and other vernacular speakers) in the courtroom and beyond. *Language* 92, 4 (2016), 948–988. https://muse.jhu.edu/pub/24/article/641206 Publisher: Linguistic Society of America.

[27] Rachael Tatman. 2017. Gender and Dialect Bias in YouTube's Automatic Captions. In *Proceedings of the First ACL Workshop on Ethics in Natural Language Processing*, Dirk Hovy, Shannon Spruit, Margaret Mitchell, Emily M. Bender, Michael Strube, and Hanna Wallach (Eds.). Association for Computational Linguistics, Valencia, Spain, 53–59. doi:10.18653/v1/W17-1606

[28] Alicia Beckford Wassink, Cady Gansen, and Isabel Bartholomew. 2022. Uneven success: automatic speech recognition and ethnicity-related dialects. *Speech Communication* 140 (May 2022), 50–70. doi:10.1016/j.specom.2022.03.009

A Questionnaire Design

A.1 Complete Questionnaire

The following presents the complete questionnaire administered to participants. Question types are indicated as follows: [O] = Open-ended response; [MC] = Multiple choice; [MA] = Multiple selection (select all that apply); [Likert] = 5-point Likert scale. The Atlanta site used an earlier version of the survey with some different response options, as noted below.

Demographics and Speech Background

Q1 [O]: How would you describe the way that you speak?

Q2 [O]: Would you be willing to share how you self-identify in terms of ethnicity or cultural background? If so, what term(s) do you use?

Q3 [MC]: Please indicate your age range: *Under 30 / 30–50 / Over 50*

Q4 [O]: Do you have any speech challenges (impairments or disabilities) e.g. stuttering, stammering, slurring, etc. that might affect your ability to use voice recognition technologies effectively? Please describe.

Voice Technology Use

Q5 [MA]: You are now going to see some different types of Small and Large Vocabulary technologies. Please select the ones that you use or have used. *Options: Stand-alone Smart Speakers (e.g., Amazon Echo, Google Home, Siri HomePod); Virtual Assistants on Mobile Phones, Tablets or Smart TVs; Voice Commands in Smart Home Appliances; Voice Commands in Car Infotainment Systems; Voice-controlled Gaming Devices; Voice Biometrics for Authentication; Voice-enabled Wearable Devices*

Q6a [O]: If you selected any of the Small or Large Vocabulary uses above: What are some tasks you use this tech for?

Q6b [O]: If there are any other tasks you use your Small or Large Vocabulary devices for, please specify.

Q7 [MA]: You are now going to see some different types of Conversational technologies. Please select the ones that you use or have used. *Options: Video conferencing apps with auto-generated transcription (e.g., captions and transcripts on Zoom, Microsoft Teams, etc.); Voice-to-text/dictation features; Automated phone systems (IVR)*

Q8a [O]: If you selected any of the Conversational uses above: What are some tasks you use your tech for?

Q8b [O]: If there are any other tasks you use your Conversational devices for, please specify.

Q9 [O]: If there are any other kinds of voice recognition technologies you have used, please specify.

Q10 [O]: Are there any devices from the list we just talked about that you used, but stopped using? Can you tell us about what led you to stopping?

Experiences with Voice Recognition

Q11 [O]: Can you tell us about your most memorable experience with voice recognition technology? What was the technology and how did you use it? Did you accomplish what you wanted through using it?

Q12a–f [Likert]: How often do you use voice recognition technology for the following purposes? *Scale: 1 = Very frequently to 5 = Never.* (a) To be productive and manage tasks; (b) Using voice commands as shortcuts for accessing apps on my device; (c) To keep up with information, news, weather; (d) To play around and discover new capabilities of voice technology; (e) To support me in accessibility (captioning speech); (f) Other (please specify).

Dialect and Accent Challenges

Q13 [O]: Can you share a time where you had a challenge with your way of speaking (dialect, accent or language) being understood by voice recognition systems?

Q14 [MA]: Which, if any of the following reasons describe the challenge(s) that you identified above, or other challenges that you have experienced? Select all that apply.

Options for the Gulf Coast, Miami Beach, and Tucson: Voice technologies don't seem to take into consideration my cultural/ethnic/racial background; Voice assistants take what I say out of context and either don't work or do something I didn't want; It takes too much physical effort for me to speak clear enough to be understood; I feel self-conscious about how I speak when talking to voice technologies

Additional options for Atlanta only: Voice technologies force me to change the way I talk to be understood; Voice technologies force me to repeat my speech over and over to be understood; When voice assistants/smart devices don't recognize my speech, I still have to manually complete a task; Voice technologies don't work for my accent/dialect most or all of the time; Voice technologies don't seem useful for my day-to-day lifestyle

[Skip logic prompts for narrative elicitation]: *You indicated that voice technologies don't consider your background. Would you like to share a time this happened? / You indicated that you feel self-conscious using voice tech. Would you like to share a time you felt this way? / You indicated that sometimes you get unexpected results using voice tech. Would you like to share a time this happened?*

Q14b [O]: If there are any other challenges or issues you have experienced, please specify.

Coping Strategies

Q15a [MA]: If you have experienced challenges in using voice recognition, are there things you do to fix or deal with the challenge? Select all that apply.

Options for Gulf Coast, Miami Beach, and Tucson: Repeat what I said; Change the words or phrases I use; Slow down my speech; Enunciate more; Speak louder; Imitate the voice assistant; Get another person to control the device for me; Manually complete the task instead; I just stop using it

Options for Atlanta: Repeat what I said; Change the way that I speak/code-switch; Imitate the voice assistant; Get another person to control the device for me; I just stop using it

Q15b [O]: If you said above that there are other ways you fix or deal with challenges in using voice recognition, please specify.

Emotional Responses

Q16 [O]: How does it make you feel when your voice assistant doesn't understand your speech/makes frequent mistakes?

Expectations and Willingness

Q17 [MC]: How often do you expect voice recognition technologies to be able to understand your speech dialect or accent correctly? *Options: Always (100% every time I use my device); Often (75% — most times I use my device); Sometimes (50% — every now and again); Rarely (25% occasionally); Never (0–20%)*

Q18a [MC]: Would you be willing to provide voice samples or feedback to help improve voice recognition technologies' accuracy for different dialects and accents? *Options: Yes, I would be interested in contributing; Maybe, I would need more information; No, I would not be interested in contributing*

Q19 [MC]: Finally, would you be willing to pay for a product/service that provides improved voice recognition for diverse dialects and helps in undoing bias? If yes, which describes how much you'd be willing to pay? *Options: Not willing to pay—I feel I should be compensated; Prefer 1-time payment; < $10/month; $10–20/month; $20–30/month; More than $30/month*

Voice Sample Collection (Optional)

Q20: Would you like to contribute your voice to our database? [If yes, participants were directed to a voice recording interface.]

Q21 [O]: Please share any thoughts or comments on participating in this exercise. What questions might you have for our team regarding the study?

Generative AI Usage Statement

ChatGPT 5.2 was used to assist with grammar editing and table formatting. The authors retain full responsibility for the originality, accuracy, and integrity of the manuscript.

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